

Corrigendum to “New modified UNIFAC parameters using critically evaluated phase equilibrium data” by Jeong Won Kang, Vladimir Diky, Michael Frenkel [Fluid Phase Equilib. 388 (2015) 128-141] ¹

Jeong Won Kang^a, Vladimir Diky^{b 2}, Michael Frenkel^b

^a Department of Chemical and Biological Engineering, Korea University, 5-1 Anam-Dong, Sungbuk-Ku, Seoul 136-701, South Korea

^b Applied Chemicals and Materials Division, National Institute of Standards and Technology, Boulder, Colorado 80305-3337, USA

Our paper reporting NIST-modified UNIFAC method contained a few typographical errors, as documented by J. Gmehling [1]. Further review of the original publication and its supplementary materials revealed minor additional necessary corrections. We therefore compiled this corrigendum in order to provide a comprehensive resource of all necessary corrections.

The most important correction is transposing the R and Q columns in Supplement Table 1. The correct header should look like

No.	Sub-group Name	R_k	Q_k		Example
1	CH3	0.6325	1.0608	hexane	2 CH3, 4 CH2
...					

with all numerical values unchanged.

In addition, main group 86 was erroneously defined as 84. The corresponding line in Supplement Table 1 should be as follows:

No.	Sub-group Name	R_k	Q_k		Example
(86) C2Cl4					
179	C2Cl4	3.3810	3.5845	tetrachloroethene	1 C2Cl4

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² Corresponding author. E-mail: diky@nist.gov

All other additions and corrections below apply to the examples in Supplement Table 1.

The example for subgroup 39 should be corrected:

39	AC2N	1.0731	0.3530	2,6-dimethylpyridine	1 AC2N, 3ACH, 2 CH3
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The example for subgroup 160 should be corrected:

160	AC2HO	1.1404	0.7600	2,3-benzofuran	5 ACH, 1 AC, 1 AC2HO
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The example for subgroup 162 should be corrected:

162	c-CH-NH	0.9796	0.6240	2,6-lupetidine	1 c-CH-NH, 2 CH3, 3 c-CH2, 1 c-CH
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The example for subgroup 163 should be corrected:

163	c-C-NH	0.7521	0.3960	2,2,6,6-tetramethylpiperidine	1 c-C-NH, 4 CH3, 3 c-CH2, 1 c-C
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The example for subgroup 188 should be corrected:

188	c-CH2-NH	1.2070	0.9360	piperazine	2 c-CH2-NH, 2 c-CH2
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The example for subgroup 198 should be corrected:

198	AC2NH	0.8701	0.2120	2,5-dimethylpyrrole	1 AC2NH, 2 ACH, 2 CH3
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The example for subgroup 305 should be corrected:

305	CNO2	2.8870	2.2410	2-methyl-2-nitropropane	3 CH3, 1 CNO2
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The example for subgroup 306 must be changed:

306	ACNH	1.1849	0.7320	diphenylamine	10 ACH, 1 AC, 1 ACNH
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The example for subgroup 309 should be added:

309	CH2(O)2	0.9347	0.7080	dimethoxymethane	2 CH3, 1 CH2(O)2
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